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OPERATING SYSTEM ADVANCED WK 5

1.SCENARIO ANALYSIS:

A web server uses a file system to store and manage different types of files required to operate websites. For example, HTML pages, images, videos, CSS files, JavaScript files, and configuration files are stored in organized directories. When a user visits a website, the web server searches the file system and retrieves the required files before sending them to the user's browser. The file system also controls how files are named, accessed, modified, and protected. Different permissions can be assigned to administrators, developers, and normal users to prevent unauthorized access or accidental changes. The server may also use folders to separate website files, log files, temporary files, and backups. Regular backups are important because hardware failure, malware, or accidental deletion can cause data loss. Therefore, an efficient file system helps the web server store data securely, locate files quickly, manage storage space, and provide reliable access to website resources.

2.CONCEPT RESEARCH:

I designed a directory structure to organize files for a small web server. The main directory contains separate folders for HTML pages, CSS, JavaScript, images, documents, logs, and backups. This structure makes it easier to locate and manage files without creating confusion. For example, images are stored in the "images" folder, while website pages are stored in the "html" folder. I learned that a well-organized directory structure improves file management and saves time when accessing data. It also helps administrators apply appropriate permissions and protect important files. Overall, designing the structure showed me how file organization improves system efficiency and security.

3.TOOL PRACTICE:

WebServer/

```
|
|
|— html/
|   |— index.html
|   |— about.html
|
|
|— css/
|   |— style.css
|
```

```
|— javascript/
|   └─ script.js
|
|— images/
|   └─ logo.png
|   └─ banner.jpg
|
|— documents/
|   └─ information.pdf
|
|— logs/
|   └─ server.log
|
└─ backups/
    └─ website_backup.zip
```

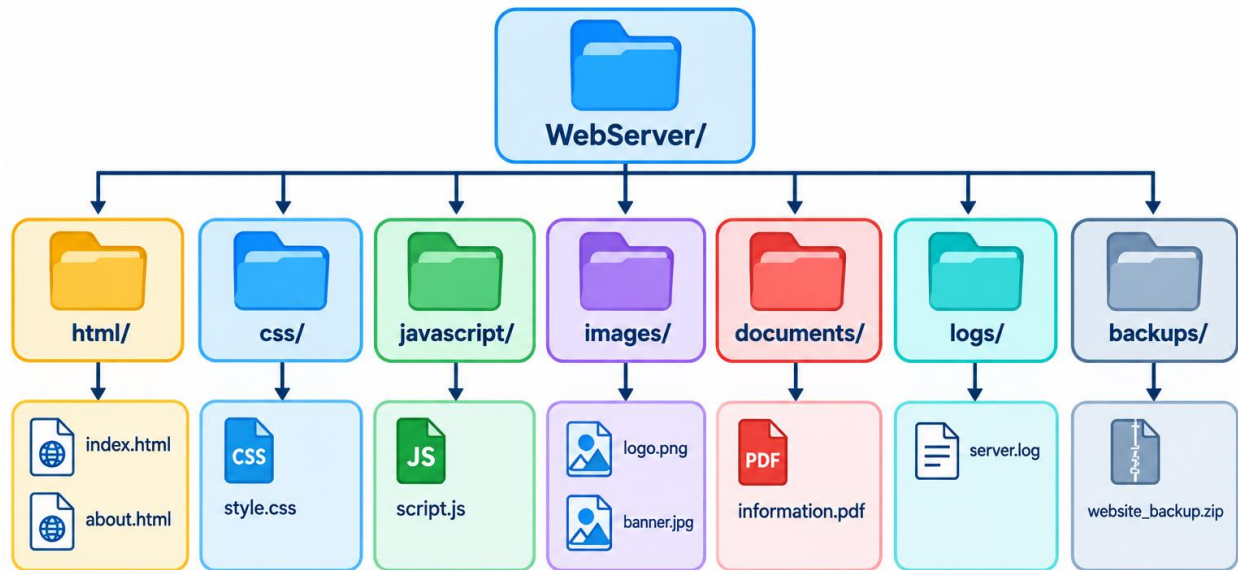
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4.DIAGRAM:



File System Diagram

Web Server Directory Structure



The file system diagram shows how files are organized in a web server. The main folder, WebServer, contains different subfolders for specific types of files. The html folder stores web pages, while the css folder contains website styling files. The javascript folder stores scripts that add functionality to the website. Images such as logos and banners are kept in the images folder. Important documents are stored in the documents folder, while server activity is recorded in the logs folder. Finally, the backups folder stores copies of important website files. This organized structure makes files easier to locate, manage, protect, and back up. It also improves efficiency and helps prevent accidental file loss.